

Caspar Popova
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Education

University of Maryland, Ph.D in Computer Science College Park, MD; 2024 - Present
Northwestern University, Bachelor of Science in Computer Science Evanston, IL; 2019 - 2022
Relevant coursework: PL + Architecture (HLS), Compilers (Garbage collection, abstract interpretation), Theorem proving (Roq, Lean), PL + Security, ML Systems, Databases

Ongoing Research

Does Rust Compile Volatile Correctly? June 2026 - Present
Testing that rustc compiles volatile code for embedded devices correctly by combining binary instrumentation testing with random program generation.
→ Random Testing, Compiler Testing, Rust, C++, Program Instrumentation

Sized Types for Program Generation May 2025 - Present
Testing recursion optimization in compilers by developing a program generator of terminating recursive programs using sized types.
→ Random Testing, Compiler Testing, OCaml, Racket, Python

Research Experience

Recursion for HLS in Panda Oct 2025 - Dec 2025
Extended the open source [Panda](#) HLS framework with support for recursion by implementing a recursion-to-iteration compiler pass.
→ Compilers, HLS, GCC, C++

PLT Group, Northwestern University Evanston, IL; 2021-2022
Designed and implemented an [SMT solver-based program synthesizer](#) for higher-order functional programs to evaluate the usefulness of error-reporting mechanisms in gradually typed languages including Typed Racket.
→ Program Synthesis, Gradual Typing, Solver-Aided Programming, Racket

Arcana Compilers Group, Northwestern University Evanston, IL; Apr - Jun 2022
Implemented a compiler transformation that lowers object abstractions and performs stack vs. heap allocation analysis as part of the [MemOIR compiler](#), which provides an intermediate representation for objects to enable specialized transformations like dead field elimination.
→ Compilers, LLVM, C++

Work Experience

Sandia National Laboratories

Remote/CA; June 2026 - Aug 2026

Graduate Research Intern

Working with John Bender to test whether the Rust compiler compiles volatile correctly.

PLUM Lab, University of Maryland

College Park, MD; Aug 2024 - Present

Graduate Research Assistant

Working with David Van Horn and Leonidas Lampropoulos on compiler testing.

Epic Systems

Verona, WI; 2022 - 2024

Software Developer in Population Health Analytics

- Drove multi-year strategic direction of an analytics framework via client feedback and collaboration with other analytics teams.
 - Built population health data models and visualizations within Microsoft SQL EDW in collaboration with multiple teams.
 - Designed internal processes and utilities to improve best practices, development, testing completeness, and coordination with other teams.
- Database Management, Data Models, Analytics, SQL, C#, TypeScript, Dashboard Design

Talks

- [Does Rust Compile Volatile Correctly?](#) SSNL 2026 (SPLASH-ISSTA)
- [Compiler Testing with Sized Types](#) NJPLS December 2025

Awards & Honors

- 2023 CSGrad4US Fellow (NSF-sponsored fellowship similar to the GRFP)

Service & Teaching

- Teaching assistant for Design and Implementation of Programming Languages, Organization of Programming Languages (OCaml, Rust, Racket) in Fall 2024 - Spring 2025.
- Student volunteer at POPL 2025.
- Peer mentoring new Ph.D. and M.S. students at University of Maryland.
- Mentoring new developers at Epic Systems.